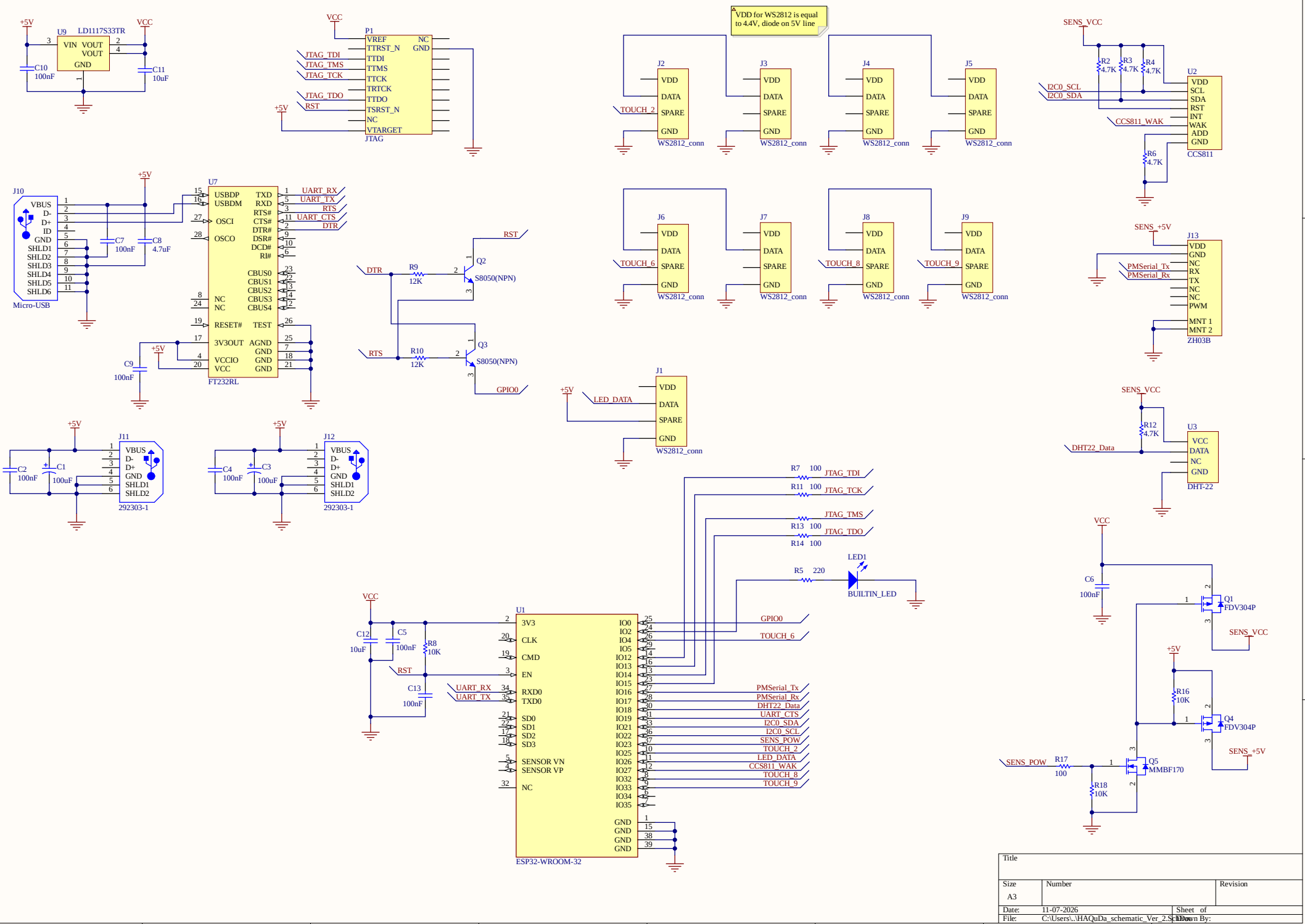
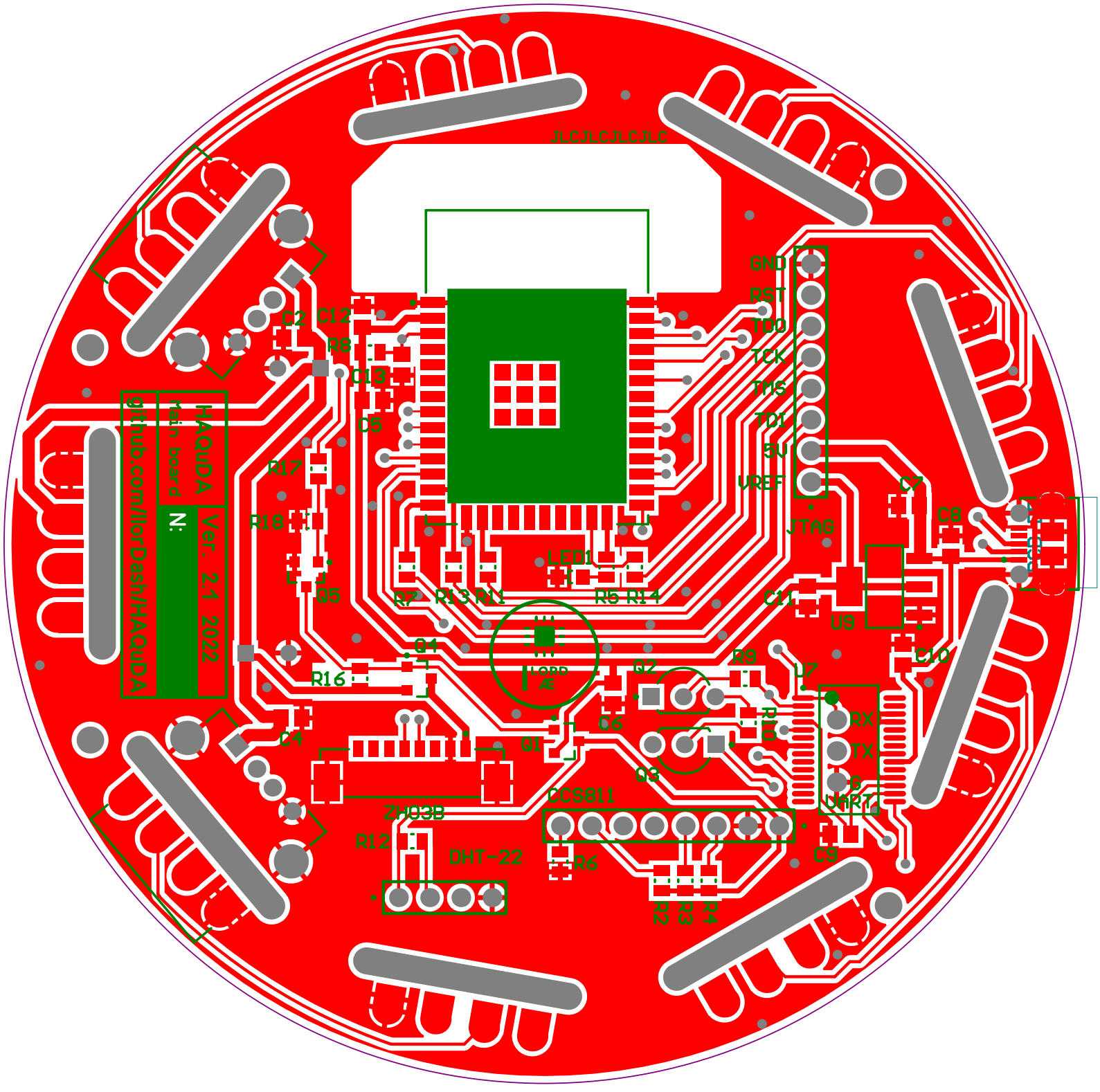
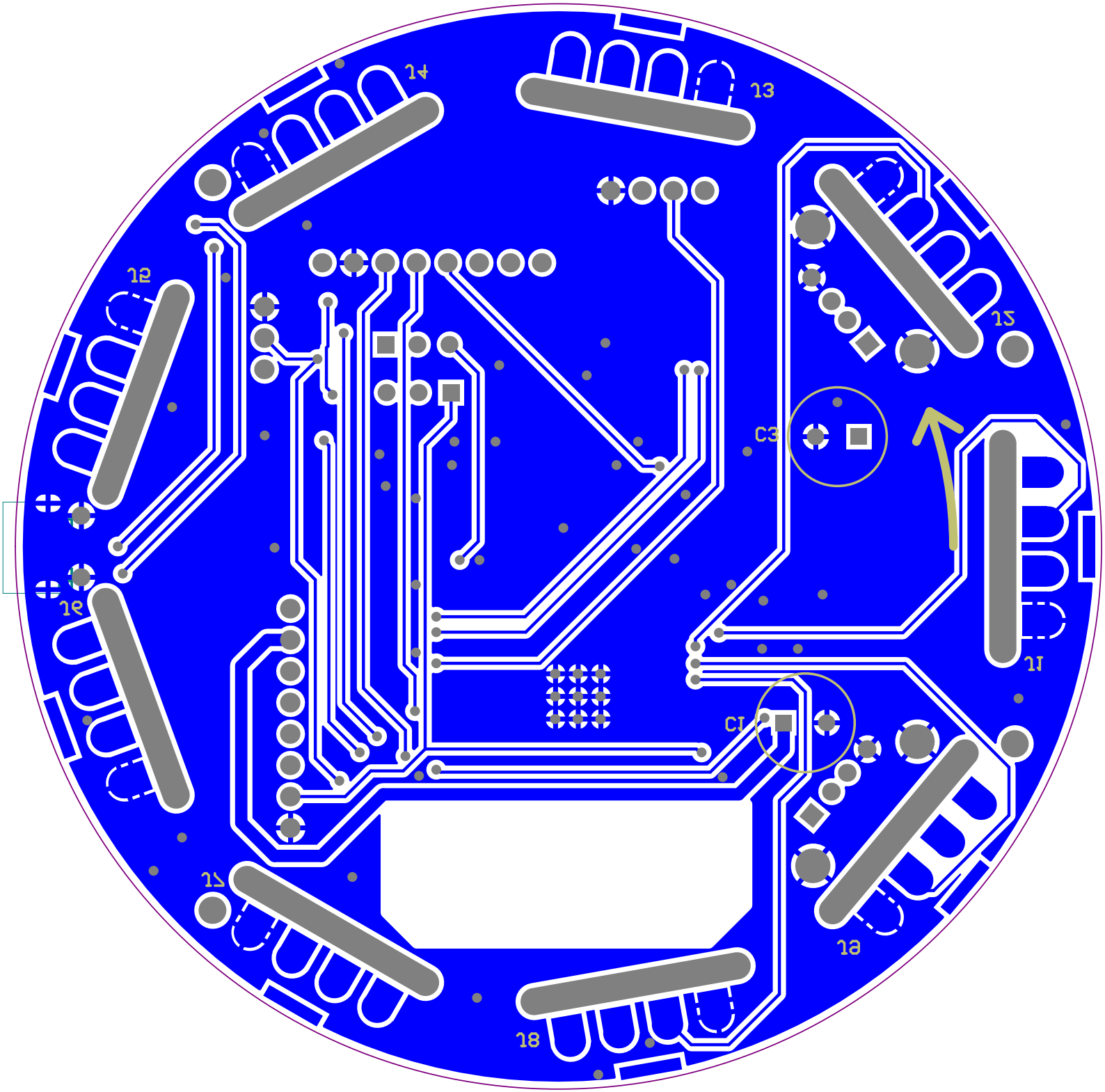

VDD for WS2812 is equal to 4.4V, diode on 5V line
U9 LD1117S33TR
VIN VOUT GND
VOUT
C10 100nF
C11 10uF
P1
JTAG
VREF TRST_N TTDI TTMS TTCK TRTCK TTDO TSRST_N NC VTARGET
NC GND
JTAG_TDI JTAG_TMS JTAG_TCK JTAG_TDO RST
J2 J3 J4 J5
VDD DATA SPARE GND
TOUCH_2
WS2812_conn
J6 J7 J8 J9
TOUCH_6 TOUCH_8 TOUCH_9
U2
CCS811
VDD SCL SDA RST INT WAK ADD GND
I2C0_SCL I2C0_SDA CCS811_WAK
R2 4.7K R3 4.7K R4 4.7K R6 4.7K
SENS_VCC
J10
Micro-USB
VBUS D- D+ ID GND SHLD1 SHLD2 SHLD3 SHLD4 SHLD5 SHLD6
C7 100nF C8 4.7uF
U7
FT232RL
USBDP USBDM OSCI OSCO RESET# 3V3OUT VCCIO VCC NC NC TXD RXD RTS# CTS# DTR# DSR# DCD# RI# CBUS0 CBUS1 CBUS2 CBUS3 CBUS4 TEST AGND GND GND GND
UART_RX UART_TX RTS UART_CTS DTR
C9 100nF
R9 12K R10 12K
Q2 Q3 S8050(NPN)
DTR RTS RST GPIO0
J13
ZH03B
SENS_+5V
PMSerial_Tx PMSerial_Rx
VDD GND NC RX TX NC NC PWM MNT 1 MNT 2
J1
WS2812_conn
LED_DATA
J11 J12
292303-1
C2 100nF C1 100uF C4 100nF C3 100uF
U3
DHT-22
VCC DATA NC GND
DHT22_Data
R12 4.7K
U1
ESP32-WROOM-32
3V3 CLK CMD EN RXD0 TXD0 SD0 SD1 SD2 SD3 SENSOR VN SENSOR VP NC
IO0 IO2 IO4 IO5 IO12 IO13 IO14 IO15 IO16 IO17 IO18 IO19 IO21 IO22 IO23 IO25 IO26 IO27 IO32 IO33 IO34 IO35
GND GND GND GND
C12 10uF C5 100nF C13 100nF R8 10K
R7 100 R11 100 R13 100 R14 100
JTAG_TDI JTAG_TCK JTAG_TMS JTAG_TDO
R5 220
LED1
BUILTIN_LED
GPIO0 TOUCH_6 PMSerial_Tx PMSerial_Rx DHT22_Data UART_CTS I2C0_SDA I2C0_SCL SENS_POW TOUCH_2 LED_DATA CCS811_WAK TOUCH_8 TOUCH_9
UART_RX UART_TX
C6 100nF
Q1 Q4 FDV304P
Q5 MMBF170
R16 10K R17 100 R18 10K
SENS_POW SENS_VCC SENS_+5V
Title
Size A3
Number
Revision
Date: 11-07-2026
Sheet of
File: C:\Users\..\HAQuDa_schematic_Ver_2.SchDoc
Drawn By:


JLCJLCJLCJLC
HAQuDA
Main board
github.com/IlorDash/HAQuDA
Ver. 2.1 2022
GND
RST
TDO
TCK
TMS
TDI
3V
VREF
JTAG
C2
C12
R8
C13
C5
R17
R18
Q5
LED1
R7
R13
R11
R5
R14
C11
U9
C7
C8
C10
Q4
R16
C4
Q1
Q2
R3
R9
U7
C6
Q3
RX
TX
G
UART
CCS811
ZH03B
R12
DHT-22
R6
C9
R2
R3
R4


J4
J3
J2
J5
C3
J1
C1
J8
J7
J6